


# **L12a**: How Do Micro Decisions Lead to Emergent Macro Trends?

CHEME 5660 Cornell University

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# Objectives for Today

Today, we write down simple rules for individual traders, simulate many of them together, and measure what the market as a whole does.

## Today's concepts

- Agent-based models: the five parts every model shares, in the minority game and the Farmer–Joshi market.
- The Boltzmann choice rule: derived from maximum entropy, with a temperature that sets how strongly agents favor high scores.
- Micro to macro: volatility in the minority game; fat tails and volatility clustering in the Farmer–Joshi market.

# What You Need to Know

## Core concepts

- Five parts: agents, information, a decision rule, an aggregate, and feedback from the aggregate to the next decisions.
- Maximum entropy for a fixed expected score gives $\pi_s = \exp(U_s/T)/\mathcal{Z}$: greedy as $T \to 0$, uniform as $T \to \infty$.
- The minority game's volatility per agent depends on $\alpha$ and $T$; Farmer–Joshi value investors and trend followers together produce fat tails and volatility clustering.

## **For understanding** (you won't reproduce these)

- The critical value $\alpha_c \approx 0.3374$ (cited, not derived) and $-d\mathcal{H}/d\bar{U} = 1/T$.
- The Farmer–Joshi threshold rules and our parameter values.

# Lectures and Examples

**Lecture:**

CHEME-5660-L12a-Lecture-AgentBasedMarkets-Fall-2026.ipynb

**Examples:**

CHEME-5660-L12a-Example-ThermalMinorityGame-Fall-2026.ipynb

CHEME-5660-L12a-Example-FarmerJoshi-Fall-2026.ipynb

# In the News: The November 5 Sell-Off

On Thursday, November 5, 2026, the Nasdaq fell 5.5% and the S&P 500 about 4.4%, the steepest one-day declines since April 2025, on worries that AI data-center spending could slow:

- The seven largest technology companies: lost more than 1 trillion USD of market value.
- Retail investors: sold a net 338 million USD of individual stocks, the largest one-day outflow since March 2020. “On a typical sell-off, retail buy the dip. Today, that bid has effectively disappeared from individual names.”

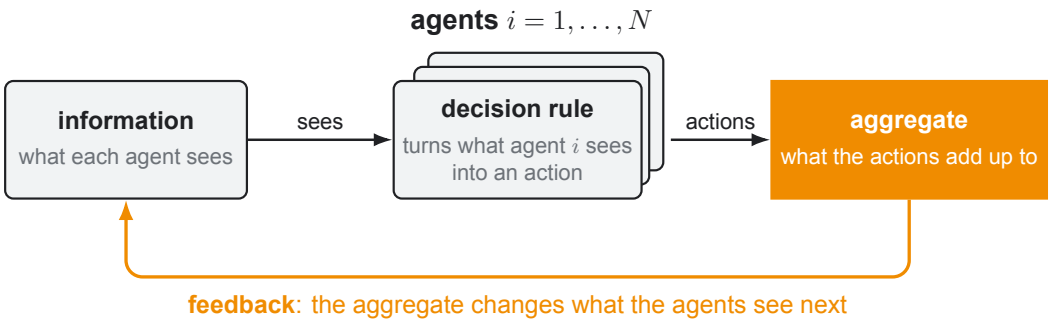
**Connection:** dip buyers act like value investors. What happens to price moves when they step back?

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**Source:** CNBC; CNBC, Vanda Research data

# Five Parts of Every Agent-Based Model

Agents see information, their decision rules turn it into actions, the actions add up to an aggregate, and the aggregate feeds back into what they see next:



# The Structure of an Agent-Based Model

An agent-based model describes individual decision makers and their rules, simulates them together, and measures the market-wide result:

| Part | Minority game | Farmer–Joshi market |
|---|---|---|
| Agents | $N$ agents, $S$ fixed strategies each | Value investors, trend followers |
| Information | History $\mu(t)$ of the last $m$ outcomes | Mispricing or price change over a lag |
| Decision rule | Choose a strategy by its score | Enter or exit at a threshold |
| Aggregate | Attendance $A(t)$ | Net order, then the price |
| Feedback | $A(t)$ updates scores and history | New price changes every signal |

# The Minority Game

Each round, $N$ (odd) agents buy ($+1$) or sell ($-1$); the minority side, $-\text{sign}\, A(t)$, wins. Agent $i$ plays the action $a_i(t)$ of one of its $S$ strategies, tables $a_{is}^{\mu}$ over the $P = 2^m$ histories $\mu$:

$$A(t) = \sum_{i=1}^{N} a_i(t) \qquad\qquad U_{is}(t+1) = U_{is}(t) - \frac{a_{is}^{\mu(t)} A(t)}{P}$$

where $A(t)$ is the attendance and $U_{is}(t)$ the score of strategy $s$ of agent $i$. A strategy that would have put its agent in the minority gains score.

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**Source:** Challet and Zhang (1997); scores divided by $P$ as in Challet, Marsili, and Zecchina (2000)

# Volatility per Agent

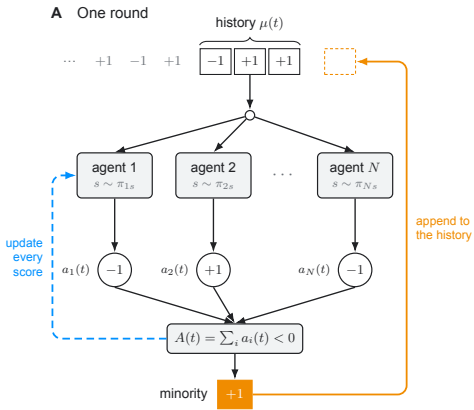
We measure how far the attendance moves from an even split, compared with $N$ agents who flip fair coins (variance $N$):

$$\frac{\sigma^2}{N} = \frac{\langle A^2 \rangle}{N} \qquad\qquad \alpha = \frac{P}{N}$$

where $\langle \cdot \rangle$ averages over rounds and $\alpha$ is the number of histories per agent. $\sigma^2/N > 1$: more agents on the same side than with coin flips; $\sigma^2/N < 1$: a more even split. When $\alpha$ is small, many agents switch to the same action at the same time.

# One Round of the Minority Game

The history goes to every agent; the actions add up to the attendance, which feeds back into the history and every strategy's score:



# One Agent's Strategies

Each strategy is a table with one action per history. The scores set the choice, and the round updates the scores:

**B** Agent $i$'s two strategies ($m = 3$)

| history $\mu$ | strategy 1 | strategy 2 | |
|---|---|---|---|
| $-1\ -1\ -1$ | $+1$ | $-1$ | |
| $-1\ -1\ +1$ | $-1$ | $-1$ | |
| $-1\ +1\ -1$ | $+1$ | $+1$ | |
| $-1\ +1\ +1$ | $-1$ | $+1$ | $\leftarrow \mu(t)$ |
| $+1\ -1\ -1$ | $-1$ | $+1$ | |
| $+1\ -1\ +1$ | $+1$ | $-1$ | |
| $+1\ +1\ -1$ | $+1$ | $+1$ | |
| $+1\ +1\ +1$ | $-1$ | $+1$ | |
| score $U_{is}(t)$ | $2.4$ | $0.9$ | |
| $\pi_{is}$ at $T = 1.5$ | $0.73$ | $0.27$ | |
| after the round | ↓ falls | ↑ rises | minority $= +1$ |

# Choosing a Strategy: Maximum Entropy

An agent chooses strategy $s$ with probability $\pi_s$. The entropy $\mathcal{H}(\pi)$ measures how spread out the choice is: zero for a certain choice, $\ln S$ for a uniform one. For a fixed expected score $\bar{U}$, we choose the $\pi_s$ that commit to the least:

$$\max_{\pi} \; \mathcal{H}(\pi) = -\sum_{s=1}^{S} \pi_s \ln \pi_s \quad \text{subject to} \quad \sum_{s=1}^{S} \pi_s = 1, \qquad \sum_{s=1}^{S} \pi_s U_s = \bar{U}$$

with $\bar{U}$ strictly between the average and the largest score.

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**Source:** Cavagna, Garrahan, Giardina, and Sherrington (1999)

# The Boltzmann Rule

With Lagrange multipliers $\phi_0$ (normalization) and $\phi_1$ (expected score), setting $\partial \mathcal{L} / \partial \pi_s = 0$ gives:

$$-\ln \pi_s - 1 - \phi_0 + \phi_1 U_s = 0 \quad \Longrightarrow \quad \boxed{\pi_s = \frac{1}{\mathcal{Z}} \exp\left(\frac{U_s}{T}\right)}, \qquad \mathcal{Z} = \sum_{k=1}^{S} \exp\left(\frac{U_k}{T}\right)$$

where $T = 1/\phi_1$ is the temperature, in the units of the scores, and the normalizing sum $\mathcal{Z}$ is the partition function. The second derivative, $-1/\pi_s$, is negative, so this is a maximum.

# What the Temperature Does

The temperature sets how much the choice depends on the scores:

- $T \to 0$: the best strategy has probability 1, the greedy minority game; with one best strategy, the entropy goes to zero.
- $T \to \infty$: every strategy has probability $1/S$; the entropy is $\ln S$.
- In between: with two scores $\Delta$ apart, the better strategy is played with probability $e^{\Delta/T}/(e^{\Delta/T}+1)$, about 0.73 at $T = \Delta$.

**Physics:** $-d\mathcal{H}/d\bar{U} = \phi_1 = 1/T$, the way thermodynamics defines temperature through entropy and energy.

# The Thermal Minority Game Algorithm

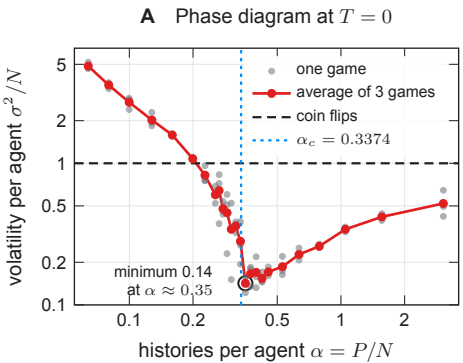
**Initialize:** $N$ (odd), $m$, $S$, $T$, rounds $R$; $P = 2^m$; random tables $a_{is}^{\mu}$; scores $U_{is}(1) \leftarrow 0$; random history $\mu(1)$. For each round $t = 1, \dots, R$ **do**:

1. Choose: draw $s_i(t)$ with $\pi_{is}(t) = \exp(U_{is}(t)/T)/\mathcal{Z}_i(t)$.
2. Act: $a_i(t) \leftarrow a_{i\,s_i(t)}^{\mu(t)}$.
3. Aggregate: $A(t) \leftarrow \sum_i a_i(t)$; the minority action is $-\text{sign}\, A(t)$.
4. Score: $U_{is}(t+1) \leftarrow U_{is}(t) - a_{is}^{\mu(t)} A(t)/P$ for every $i$ and $s$.
5. History: drop the oldest action in $\mu(t)$, append $-\text{sign}\, A(t)$.

**Output:** $A(1), \dots, A(R)$ and $\sigma^2/N = \langle A^2 \rangle / N$ after a burn-in.

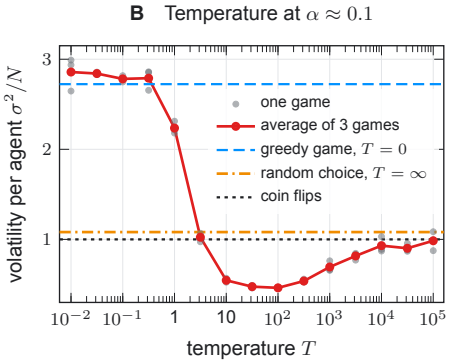
# The Phase Diagram

Volatility per agent against $\alpha$ at $T = 0$ ($m = 6$, $S = 2$, three games per $N$). The minimum, 0.14 at $\alpha \approx 0.35$, is close to $\alpha_c \approx 0.3374$:



# The Effect of Temperature

At $\alpha \approx 0.1$: greedy 2.72, random choice 1.08, and 0.46 at $T = 100$, below both limits. Our scores are divided by $P$, so our $T$ is $1/\alpha$ times that of Cavagna and coauthors:



# The Farmer–Joshi Market

Trader $i$ holds $x_t^{(i)} \in \{-c_i, 0, +c_i\}$ and orders $\omega_{t+1}^{(i)} = x_{t+1}^{(i)} - x_t^{(i)}$. The market maker moves the log price in proportion to the net order:

$$p_{t+1} = p_t + \frac{1}{\lambda} \sum_i \omega_{t+1}^{(i)} + \xi_{t+1}$$

where $\lambda$ is the liquidity and $\xi_{t+1}$ is Normal noise with standard deviation $\sigma_\xi$. The log return is $r_{t+1} = p_{t+1} - p_t$.

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**Source:** Farmer and Joshi (2002)

# Value Investors and Trend Followers

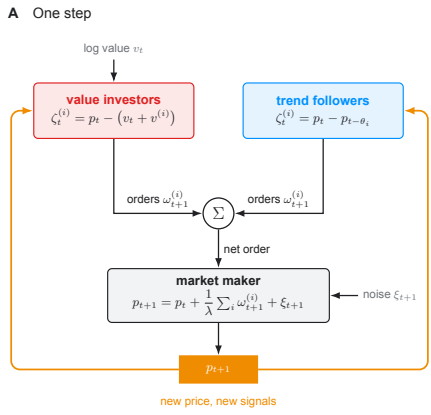
Each trader enters at a signal beyond $\pm\chi_i^{\text{in}}$ and exits back across $\pm\chi_i^{\text{out}}$, with position size $c_i = a(\chi_i^{\text{in}} - \chi_i^{\text{out}})$:

- Value investors: signal $p_t - (v_t + v^{(i)})$; sell when overpriced, buy when underpriced.
- Trend followers: signal $p_t - p_{t-\theta_i}$; buy after a rise, sell after a fall.

**Our variant:** Table 1 traders, 1,200 of each, with $\sigma_\xi = 0.05$ (Table 1 lists 0.35, which lets the price drift far from the value) and a random-walk log value $v_t$.

# One Step of the Market

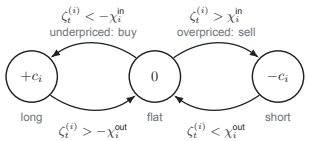
Signals set positions, positions set orders, and the net order sets the price, which changes every signal at the next step:



# The Threshold Rule

A value investor's position is one of three states. It enters when the mispricing passes $\pm\chi_i^{\text{in}}$ and exits when it comes back past $\pm\chi_i^{\text{out}}$:

**B** Threshold rule of value investor $i$




Otherwise the position stays the same.

Position size: $c_i = a\left(\chi_i^{\text{in}} - \chi_i^{\text{out}}\right)$

Trend followers use the same rule with the signs reversed: buy when $\zeta_t^{(i)} > \chi_i^{\text{in}}$, sell when $\zeta_t^{(i)} < -\chi_i^{\text{in}}$.

# The Farmer–Joshi Algorithm

**Initialize:** thresholds $\chi_i^{\text{in}} > \chi_i^{\text{out}}$, offsets $v^{(i)}$, lags $\theta_i$, $a$, $\lambda$, $\sigma_\xi$, $\sigma_\eta$, steps $R$; $c_i \leftarrow a(\chi_i^{\text{in}} - \chi_i^{\text{out}})$; $x_0^{(i)} \leftarrow 0$. For each step $t = 0, \dots, R-1$ **do**:

1. Signals: $\zeta_t^{(i)} \leftarrow p_t - (v_t + v^{(i)})$ (value) or $p_t - p_{t-\theta_i}$ (trend, once $t \geq \theta_i$).
2. Positions: the threshold rule sets $x_{t+1}^{(i)} \in \{-c_i, 0, +c_i\}$.
3. Orders: $\omega_{t+1}^{(i)} \leftarrow x_{t+1}^{(i)} - x_t^{(i)}$.
4. Price: $p_{t+1} \leftarrow p_t + \frac{1}{\lambda} \sum_i \omega_{t+1}^{(i)} + \xi_{t+1}$.
5. Value: $v_{t+1} \leftarrow v_t + \eta_{t+1}$.

**Output:** log prices, log values, log returns $r_{t+1} = p_{t+1} - p_t$, volume.

# Stylized Facts: Simulated and Real

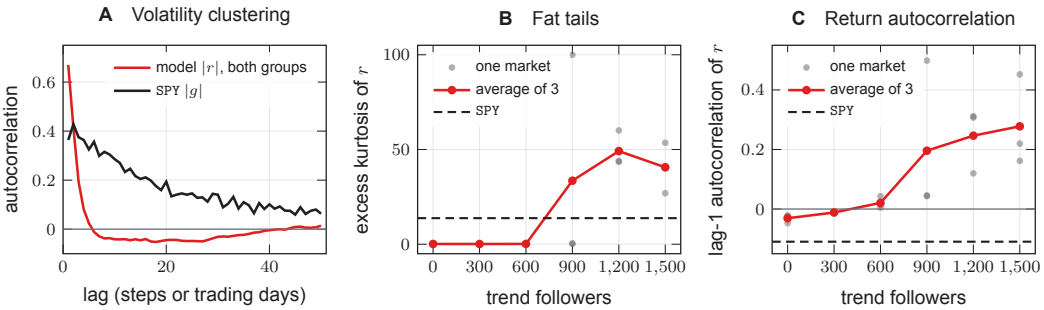
Both groups together, against `SPY` daily growth rates, 2014–2024 (kurtosis and autocorrelations do not depend on units):

| Statistic | Simulated | `SPY` |
|---|---|---|
| Excess kurtosis | 45.3 | 13.8 |
| Autocorrelation of $r$, lag 1 | 0.27 | $-0.11$ |
| Autocorrelation of $\lvert r\rvert$, lag 1 | 0.67 | 0.36 |
| Autocorrelation of $\lvert r\rvert$, lag 5 | 0.03 | 0.33 |
| Autocorrelation of volume, lag 1 | 0.56 | 0.75 |

Fat tails and clustering come out of the rules, but the clustering fades within a few steps.

# Clustering and the Mix of Traders

The model's clustering fades within a few steps (A). Fat tails and a positive return autocorrelation appear only with enough trend followers (B, C); every mix misses SPY by a wide margin on at least one statistic:



# Summary

We introduced agent-based models, derived the Boltzmann choice rule from maximum entropy, and connected trading rules to market-wide behavior.

## Key takeaways

- Agent-based models: agents, information, a decision rule, an aggregate, and feedback.
- Maximum entropy: gives $\pi_s \propto \exp(U_s/T)$; the multiplier on the expected score is $1/T$.
- Micro to macro: the minority game's volatility depends on $\alpha$ and $T$; Farmer–Joshi traders produce fat tails and clustering, but not every stylized fact at once.